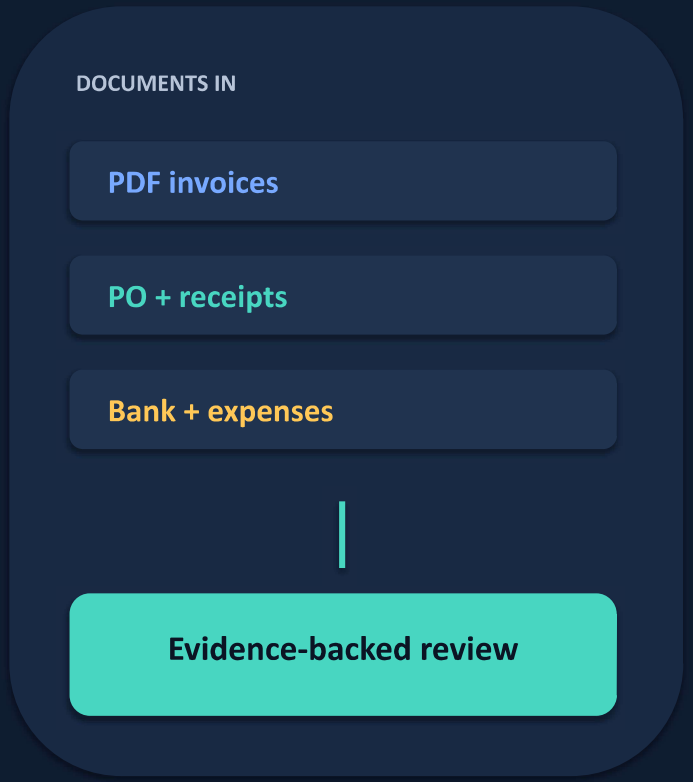
CAPSTONE SOLUTION

Invoice & Expense Reconciliation

From mixed documents to defensible finance decisions.



DEMO: UPLOAD | INSPECT | REVIEW | DRAFT

01 / WHY THIS MATTERS

Finance teams reconcile evidence scattered across formats

The solution aims to reduce manual comparison while keeping decisions reviewable.

Fragmented inputs

Invoices arrive as PDFs and scans; purchase orders, receipts, and statements may be CSVs.

Exception risk

Price, quantity, tax, duplicate, payment, and policy issues are easy to miss in manual review.

Auditability

A useful outcome needs traceable evidence, expected-versus-found values, and a recorded human disposition.

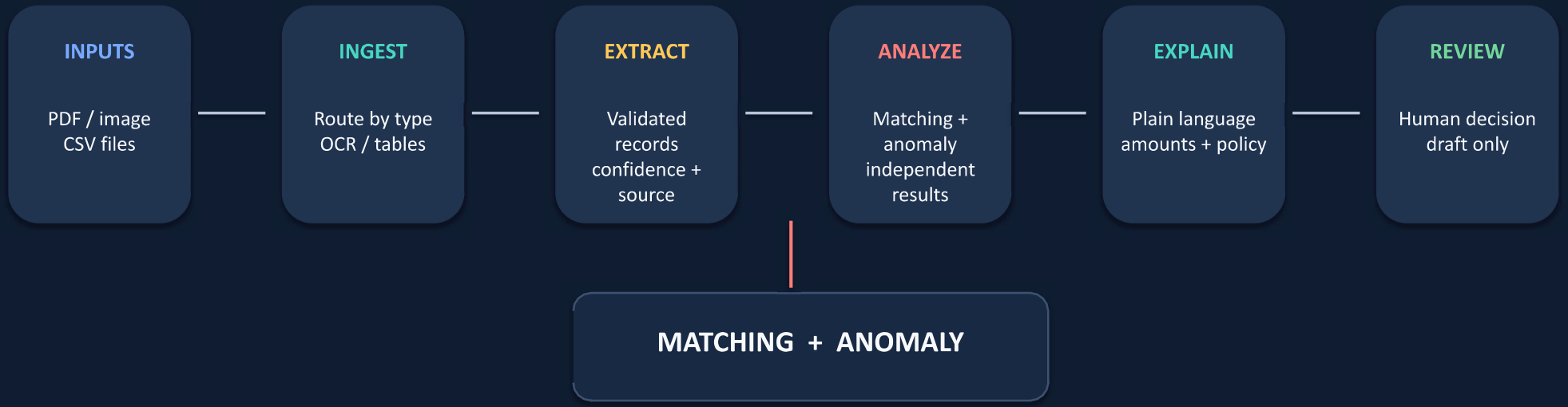
DESIGN GOAL

Automate document handling and repeatable checks; make exceptions easy to understand; keep approval authority with people.

02 / SOLUTION OVERVIEW

A specialist-agent pipeline; deterministic checks stay deterministic

Matching and anomaly detection are separate analysis paths; the Explanation Agent consumes both.



Outputs remain distinct by stage and are inspectable in the browser demo.

Each agent has a clear job and a clear boundary

Use language models where interpretation helps; use code where repeatability and auditability matter.

Ingestion

PDF text layer, OCR for scans, CSV tables; source tagging.

Extraction

Structured Pydantic records; missing fields, varied formats, field confidence.

Matching

Deterministic PO / invoice / receipt / payment checks with tolerances.

Anomaly

Independent duplicate, vendor-history outlier, and policy-limit checks.

Explanation

Consumes matching + anomaly results; policy retrieval; optional LLM wording.

HITL + Correspondence

Routes cases, records decisions, prepares vendor drafts for review.

Matching and anomaly answer different questions

Keep their findings separate; let the Explanation Agent combine the context for humans.

MATCHING: does the transaction agree?

Invoice vs purchase order
Invoice vs receipt / service entry
Bank debit vs invoice amount

Example: expected unit price INR 9,912; found INR 10,512.

ANOMALY: is this invoice unusual or risky?

Duplicate invoice identity / content
Amount outlier vs vendor history
Above configured policy limit

Example: INR 110,448 exceeds a INR 100,000 policy threshold.



Independent findings can coexist; the case should preserve each source and avoid hiding duplicate signals.

Ground the explanation; never let the model invent the numbers

The current demo uses deterministic local policy retrieval and optional Azure OpenAI generation.

1 Facts

Verified issue text, expected / found amounts, currency, severity, evidence, and duplicate reference.

2 Policy retrieval

Keyword-ranked passages from local Markdown policy files; citations stay attached to output.

3 Wording

Optional LLM generates a concise explanation and action. Deterministic wording is the fallback.

GUARDRAIL

Amounts and evidence are sourced from deterministic results. The model shapes wording; it does not calculate or replace the verified values.

LLM explanations are disabled by default; enabling them sends issue context and retrieved policy passages to the configured Azure endpoint.

06 / USER EXPERIENCE

One local UI for upload, stage inspection, and review

FastAPI serves the browser UI and endpoints; an operator uploads one scenario bundle per run.

01

Upload

PDFs, scans, and CSVs. Files are stored under a run-specific folder.

02

Inspect

Open ingestion, extraction, matching, anomaly, and explanation JSON tabs.

03

Review

See findings, evidence, confidence, amount at risk, and routing reasons.

04

Decide / draft

Record reviewer decision; prepare a vendor email draft for human review.

Demo limits: 50 files / run; 25 MB per file; 100 MB total. Descriptive names help classify uploaded documents.

07 / HUMAN CONTROL

The system recommends and records; it does not authorize payment

High-risk or uncertain cases route to people. Communication remains draft-only.

Human review routing

Matching exceptions, anomalies, low confidence, and amounts at/above the auto-clear limit enter the review queue. Decisions persist in SQLite.

Explicit safety boundaries

Approve / reject are review dispositions only. No payment release. Vendor correspondence is prepared as a draft; no email is sent.

DEMO DEFAULTS

Minimum extraction confidence: 0.90 | Auto-clear amount ceiling: 50,000 (currency units)

Production follow-up: authentication, role-based access, retention controls, approval policy ownership, and operational monitoring.

Four compact scenarios make the workflow easy to present

The sample bundle is in artifacts/reconciliation-input-use-cases.zip; use one folder at a time.

01	**Clean match**	Pass path through invoice / PO / receipt checks
02	**Tax exception**	Tax-type / rate issues become reviewable findings
03	**Duplicate invoice**	Matching and anomaly engines both surface duplicate signals
04	**Policy + price**	Price variance plus amount-above-limit anomaly

Run: start uvicorn, open localhost:8000, upload all files from one scenario, then inspect each stage tab.

09 / WHAT IS READY

A demonstrable end-to-end capstone; production hardening is next

A working vertical slice connects document processing to exception handling and human review.

Implemented

Upload UI + FastAPI
Ingestion and extraction
Deterministic matching + anomaly checks
Grounded explanations + review queue
Draft-only correspondence

Next to mature

Stable report/run correlation
Group duplicate signals without losing provenance
Richer anomaly comparison evidence
Access control, retention, deployment, monitoring
Representative evaluation and thresholds

Core principle: automate repetitive checks, preserve evidence, and keep consequential decisions with people.

Full OCR demo needs Tesseract; unstructured extraction needs configured OpenAI credentials.